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Microcontrollers:

MCUs Play A Major Role... In Terms Of INTELLIGENCE At A Higher Level

What is driving more engineers to use modern microcontrollers (MCUs) for their latest designs? This interview takes a look at the little things about MCUs. Vishal Malhotra, national sales head - automotive, Infineon Technologies India, and Andy Wong, senior regional marketing manager for Industrial Microcontroller (IMC), PMM division, Infineon Singapore APAC, speak with Dilin Anand from EFY

connectivity on demand will be the key considerations.

As for automotive, the most important feature required is secured connectivity.

Q. How are firms like yours stepping up to the security game?

A. XMC4000 Safety Package is an example of the trend going forward. Users receive not just MCU hardware but also detailed documentation and certified software test library, including consultancy and implementation support by an embedded engineering tool supplier.

Q. What are the prime features being driven by modern MCUs?

A. MCUs play a major role not only in terms of speed and performance but also in terms of intelligence at a higher level. Introduction of smart features on the interior and exterior of cars is driving the need for faster processors. MCUs with high data-processing capabilities are in demand. Driving this demand further is the concept of connected cars.

Q. What are the defining factors separating technology advances in CPUs versus those in MCUs?

A. Unlike CPUs, which are driven by advancing technology nodes, MCUs with their peripherals and multi-functional capabilities have to address a wide range of applications used in different industries.

For example, AURIX MCU for automotive has ten cores, four of which are core-checkers to ensure that the six multi-cores are properly working at all times in the harsh

automotive environment.

Q. How are things advancing on 8-, 16- and 32-bit architectures?

A. With advances in technology and development of smaller node sizes, cost difference between a 32-bit MCU and an 8-bit or a 16-bit MCU is becoming insignificant. In my point of view, it is foreseeable that 32-bit architecture would become the mainstay architecture for MCUs of the future.

Q. What would you like to see in an MCU designed specifically for IoT applications?

A. The Internet of Things's (IoT's) key requirement is connectivity, which includes both wired and wireless. Connectivity in itself needs to consider security and real-time control. Infineon's XMC4800, for example, is an MCU with built-in etherCAT for real-time applications in Industry 4.0 space.

In the consumer space of IoT applications, ultra-low power and

Q. What benefits do software bundles like this offer to engineers?

A. Modern MCUs will be more customisable, not only from a hardware standpoint but also software bundling to better address the applications in focus. XMC4000 MCU further advances the development of functional safety in industrial applications such as factory automation, industrial motor control and robotics.

Q. How are the latest chips reducing power consumption compared to those launched last year?

A. Low-power systems break into low-power active and low-power standby. Real-time high speed and low power are contradictory to one another. By developing multiple low-power domains and proper system partitioning, an MCU is likely to be more adept at meeting the demands of different IoT applications. **EFY**